

GENETICS 101 · ANSWER KEY - PROFESSOR COPY

Homework 1: The Instruction Book

Grading guide. Total 21 points. Answers in different words that mean the same thing get full credit; partial credit is suggested where it applies. Enter the final score in Professor mode on the recitation page.

Part A — Decode the message

Cells read DNA three letters at a time. Here is a pretend code table: ATG = B, CCA = A, GGT = Y, TTA = O, GAC = L, CTG = E. Message to decode: ATG CCA ATG GGT

1. Decode the message above, one three-letter word at a time. Write the word it spells. (4 pts)

Answer: B-A-B-Y -> BABY. 1 pt per correctly decoded letter.

Part B — Counting the book

2. A baby receives 23 chromosomes from its mother and 23 from its father. How many chromosomes does the baby's body cell have? (1 pt)

Answer: 46.

3. Before a body cell divides, it copies every one of its chromosomes. How many chromosomes are inside it just before it splits? (1 pt)

Answer: 92 (46×2) — then each new cell gets 46.

4. DNA has 4 letters, and cells read them in 3-letter words. How many different 3-letter words are possible? (Hint: 4 choices for the first letter $\times$ 4 for the second $\times$ 4 for the third.) (2 pts)

Answer: $4 \times 4 \times 4 = 64$. 1 pt for setting up $4 \times 4 \times 4$, 1 pt for 64.

Part C — Identical or fraternal?

5. Write IDENTICAL or FRATERNAL for each pair of twins. (4 pts)

Answer: Identical; Fraternal (a boy and a girl can't be identical — different sex chromosomes); Fraternal; Identical.

Part D — Your family tree

6. Draw a family tree: you at the bottom, your two parents above you, and your four grandparents above them. Draw an arrow from each parent to you and write “23” on each arrow to show the chromosomes you received. (2 pts)

Answer: Three generations drawn (1 pt); two arrows into the child each labeled 23 (1 pt).

7. What did Rosalind Franklin's "Photo 51" show? (1 pt)

Answer: A. The X-shaped pattern that revealed DNA is a twisted ladder (a double helix). - The X pattern in her 1952 X-ray photograph was the fingerprint of a helix.

Part E — Flashback**8. From Biology: which part of the cell holds the DNA? (1 pt)**

Answer: The nucleus.

9. From NICU Tutor and Chemistry: hemoglobin is a _____ that carries _____. (2 pts)

Answer: Protein; oxygen. 1 pt each.

Part F — Think like a doctor**10. Twins fill a lot of NICU beds. Give TWO reasons twins are more likely than single babies to need the NICU, and say which NICU Tutor machine each twin might need first. (3 pts)**

Answer: Reasons (1 pt each, any two): twins are usually born early (about 60% preterm), so their lungs, brains, and feeding aren't ready; two babies sharing one womb are often smaller; sometimes one twin gets less blood/nutrition than the other. Machine (1 pt): any sensible link — e.g., CPAP for immature lungs, isolette for a small baby who can't stay warm, feeding tube for a baby too early to suck-swallow-breathe.